

ASTROMETRIC MEASUREMENTS OF STF 1434 DOUBLE STAR SYSTEM

Fernando.C¹, Withanage N.S.D⁽¹⁾, Kalawila Vithanage. S.P⁽¹⁾, Heenatigala T⁽²⁾⁽³⁾

¹ Department of Physics, University of Ruhuna, Matara, Sri Lanka; chalakafernando4@gmail.com

² Earth-Life Science Institute (ELSI), Japan.

³ National Astronomical Observatory of Japan (NAOJ), Japan.

Abstract

A comprehensive study of the STF 1434 double star system (also known as WDS 10274+1804 according to the Washington Double Star (WDS) catalogue) was conducted utilizing the Las Cumbres Observatory (LCO) telescope located at the Cerro Tololo observatory. The target double star was selected through a rigorous process employing Stellarium software and the Stelledoppie catalogue. Observations were carried out from December 2023 to January 2024. The data reduction phase involved utilizing AstroImageJ software, enabling precise analysis of the observed data. Additionally, historical data of the STF 1434 double star system provided by the United States Naval Observatory (USNO) was used to compare with the new measurements. Key parameters such as Position Angle (PA), Angular Separation (SEP), and Delta Magnitude were measured from the observational data. Subsequently, a comparative analysis with historical data was conducted. The results of PA $282.1737^\circ \pm 0.0655^\circ$, SEP $6.4661'' \pm 0.0131''$, and Delta Magnitude 0.0427 ± 0.0002 indicated that the newest measurements of PA, SEP, and Delta Magnitude were consistent with the historical data with some exceptions during some occasions.

1. Introduction

Since the early use of telescopes, astronomers have recognized that stars are not uniformly luminous points but often appear as double or multiple-star systems. Double stars are any pairs of stars appearing close to each other, as first observed in 1650 by John Baptiste Riccioli with Mizar in the Big Dipper. These systems laid the foundation for studying double stars (Dunlop, 2012; Fraknoi et al., 2016a). A key challenge in double star research is distinguishing between optical and physical doubles. Optical doubles appear close only from Earth's perspective, while physical doubles are gravitationally bound systems (Badami et al., 2018; Karunarathne et al., 2022). The Mizar system exemplifies this complexity, forming both an optical double with Alcor and a physical binary with another star (Dunlop, 2012). Physical double stars, known as binary stars, exhibit various forms such as visual, spectroscopic, and astrometric binaries based on observational techniques (Karunarathne et al., 2022). Binary systems orbit a shared center of mass, making them significant in astrometry. Their study is vital for astrophysics, helping constrain the mass-luminosity relationship for stars (Cahill et al., 2018; Dugan et al., 2019). Astronomers determine stellar masses and validate theoretical models by analyzing orbital parameters like Position Angle (PA) and Angular Separation (SEP). Hence, this work provides crucial insights into stellar evolution theories (Eggleton, 2006; Fraknoi et al., 2016b).

The primary objective of this study is to compare the historical data with the newest observed data from the LCO 0.4m telescope with a Santa Barbara Instruments Group (SBIG) Charge Coupled Device (CCD) camera and update the double star catalogue for the advancement of the field of astrophysics. This helps amateur and professional astronomers to perform the orbital calculation, which is necessary for describing a particular star's stellar evolution. The comparison of the newest data with the historical data ensures the consistency

and accuracy of the latest data. In addition, it may reveal unexpected changes or events in double-star systems, such as the emergence of new companion stars and eruptive phenomena like novae and supernova.

In order to observe the astrometric measurements of the target double star system, 0.4m LCO telescope with a SBIG CCD camera and a Sloan Digital Sky Survey (SDSS) red filter were used. The images observed from the telescope were analyzed using AstroImageJ software.

2. Target selection

Selecting the target star consisted of many phases. Firstly, five constellations were chosen with the help of the Stellarium software. Stellarium provides versatile viewing options, allowing users to adjust the appearance of the sky by changing parameters. All the constellations were chosen to minimize the air mass effect. It ensured that all the constellations were near the zenith point. Practically, the constellations with lower zenith angles were considered as it was hard to obtain zero degree zenith angle measurements from a telescope. Also the constellations that were close to the moon were neglected as the moonlight might have saturated the CCD camera sensors.

As the second step, possible ranges of angular separation (SEP), Primary and Secondary magnitudes, delta magnitude, number of historical observations, and the constellations were set on advanced search method on the Stelle Doppie website as show in figure 1.

Date last	equal to	
Mag pri	more than	6
Mag sec	less than	11
Delta magnitude	less than	3
Separation	between	0.1,25
Spectral class	begins for	
Bayer	contains	
Flamsteed	equal to	
Constellation	equal to	
ADS	equal to	
HD	equal to	
HR	equal to	
HIP	equal to	
SAO	equal to	
Tycho2	equal to	
Gaia DR2	equal to	
Dm number	equal to	
Observations	more than	10
Notes flags	contains	
WDS Notes	contains	
Reports	more than	

Figure 1: Advanced search method on Stelle doppie

Possible stars were recorded for each constellation, and the Stelle Doppie catalogue was accessed by considering parameters such as delta magnitude, position angle, angular separation, primary magnitude, and secondary magnitudes of the stars. Then, the visibility of each star during the observational period was checked using the LCO in-built visibility calculator. By analyzing the visibility plots for all the stars, only six stars were visible during the observational period. Out of these visible stars, STF 1434 double star system was chosen as the target double star system under many criteria such as angular separation, delta magnitude etc. Angular separation was mainly considered as it was a vital part in the Rayleigh criterion. Angular separation considered to be at least 5 arc seconds and delta magnitude was considered to be less

than 3 so that can be resolved by the LCO 0.4m telescope which located in Cerro Tololo observatory in Chile.

The selected target double star (STF 1434) in Leo constellation also identified as WDS 10274+1804, was first observed back in 1830 with an angular separation of 6.08 arc seconds and a position angle of 269.6 degrees. Also, the parallaxes and the proper motion of the selected target double star were rechecked using one of the most accurate databases, the European Space Agency's (ESA) GAIA catalogue. The celestial object had not been observed since 2019; hence, the observations were carried out using the LCO 0.4m telescope with an SBIG-STL-6303 optical imager to update the Washington Double Star (WDS) catalogue and the other catalogues.

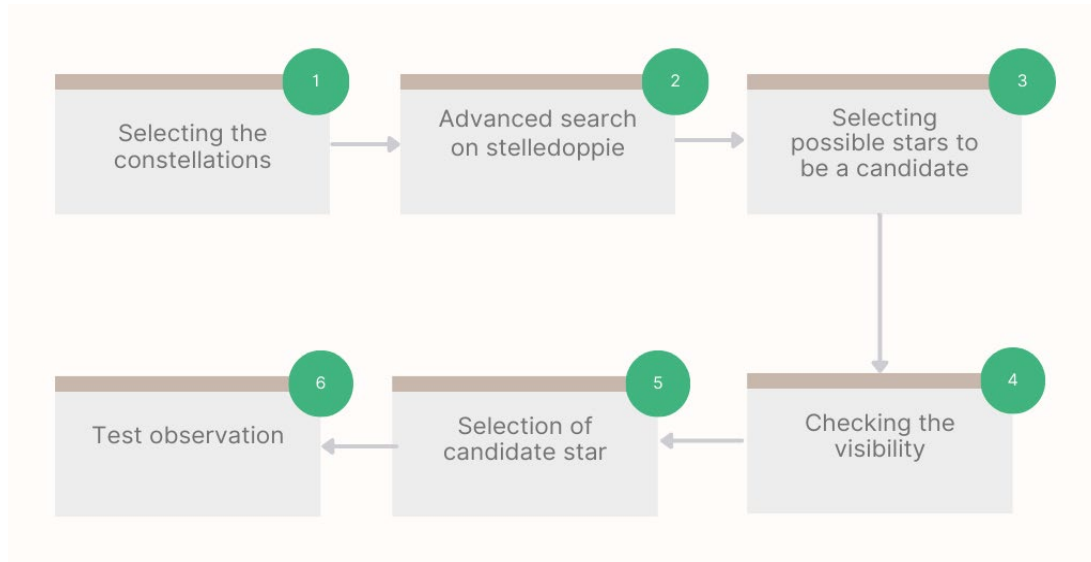


Figure 2: Schematic representation of steps in the target selection process

3. Observations

Firstly, on December 17, 2023, a test observation was carried out for all six possible stars using the above-mentioned LCO telescope. The exposure time was calculated for each star using the in-built exposure time calculator on the LCO website. Then, observations were made for each star by taking one exposure count with a different set of filters.

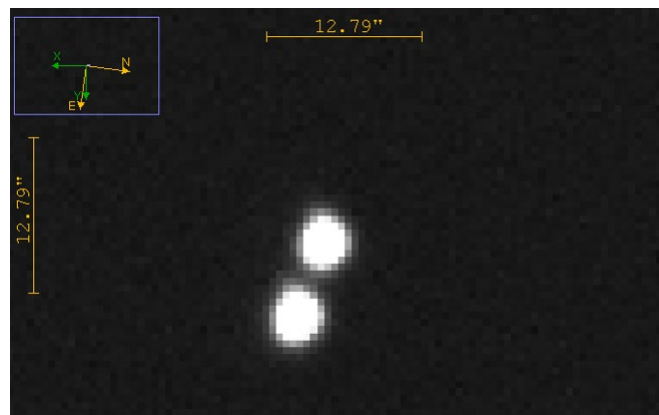


Figure 3: Test Observation for STF 1434

After the test observation, the final observation was carried out for the candidate double star system, STF 1434. The observation was made remotely using the 0.4m telescope at Cerro Tololo Observatory in Chile, which consists of an SBIG STL-6303 optical imager.

Firstly, the candidate double star system was observed with 1 s, 3 s, and 5.5 s exposure times to check whether the images were saturated or overexposed. Each time, three exposure counts were taken to use the telescope time efficiently. Flexible Image Transport System (FITS) images were visualized in AstroImageJ software. 5.5 s was selected as the optimum exposure time for the candidate double star system STF 1434.

Then, the final observation request query was made on January 17, 2024 through the LCO interface after setting the exposure time as 5.5 s and the Sloan Digital Sky Survey (SDSS) red filter as the filter. Eight exposure counts were taken to observe the STF 1434 double-star system. The observation was carried out on January 18, 2024 utilizing the Cerro Tololo observatory 0.4m telescope remotely, and the images were obtained on the same day.

Telescope Size

☐ All ☐ 2-meter ☐ 1-meter ☒ 0.4-meter

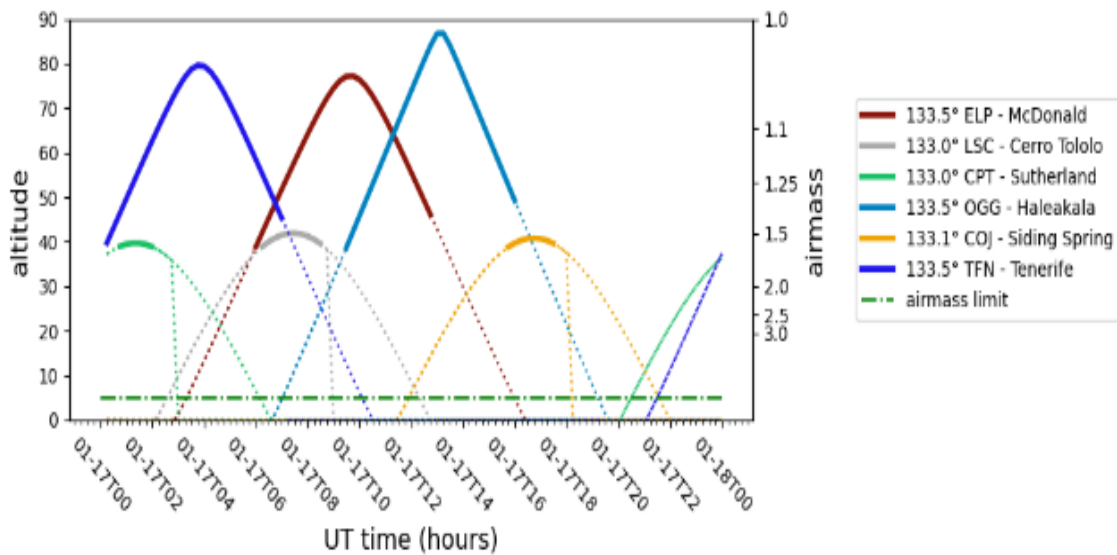


Figure 4: visibility of STF 1434 on 17th January 2024

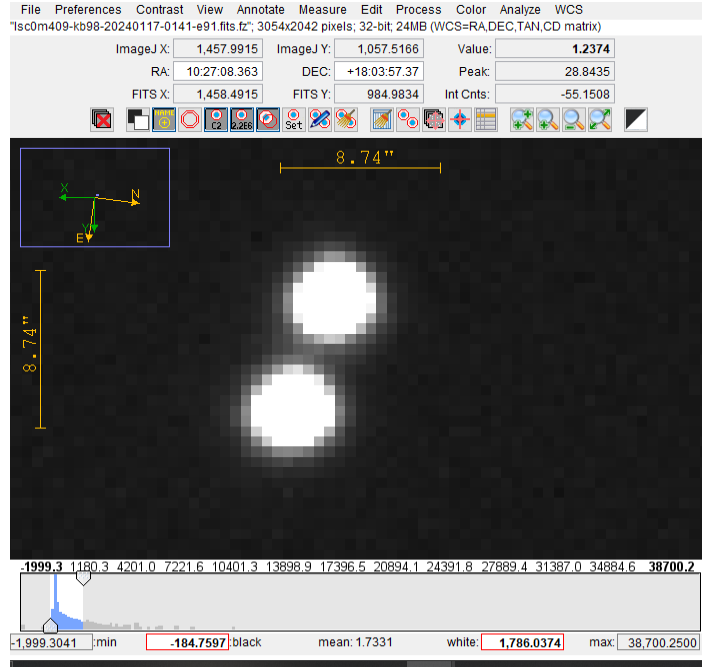


Figure 5: FITS image of STF 1434

Finally, the historical data for the specified double star system was requested from the United States Naval Observatory (USNO) which maintains the WDS.

4. Data Reduction and Data Analysis

FITS images were opened with the AstroImageJ (AIJ) software package. Then by using the View option in AIJ software, X and Y coordinates were inverted to follow the reflection of the telescope. In order to perform the multi aperture photometry, the radius of the photometric aperture, the inner background annulus and Outer background radius was set to 5, 8 and 27 pixels, respectively. When performing the multi-aperture photometry, the software package accurately identifies the centroids of the two stars (Figure 5). These centroids have specific coordinates. In the astrometry method, the position angle and the angular separation were calculated from the RA and DEC positions of the star pair according to the following equations 1 and 2.

Equation 1

$$\rho = \sqrt{(\alpha_b - \alpha_a) * \cos^2 \delta_a + (\delta_b - \delta_a)^2}$$

Equation 2

$$\theta = \tan^{-1} \left(\frac{\delta_b - \delta_a}{(\alpha_b - \alpha_a) \cos \delta_a} \right)$$

Where ρ and θ denotes SEP and PA respectively. α_a and α_b denote the RA of the brighter star and fainter star respectively and δ_a and δ_b denote the DEC of the brighter star and fainter star respectively.

According to the Equations, 1 and 2, it was necessary to know the RA and DEC of the two stars to calculate PA and SEP. Here, each image consists of pixels. Each pixel must have its own coordinates/location occupied in the sky. The process of producing the coordinates for each pixel is called as plate solving. By doing the plate solving, each pixel ensured to have its own RA and DEC in the sky. The images were already plate solved in the LCO. Therefore, the centroids of the two stars had their own RA and DEC. By connecting the centroids using the mouse, PA, and SEP were calculated from the AIJ software for one FITS image.

This process was repeated for the rest of the FITS images of the STF 1434 double star system, which were observed from the LCO telescope network. The data reduction process was performed, and the data were saved in a comma-separated value (CSV) format file for future use.

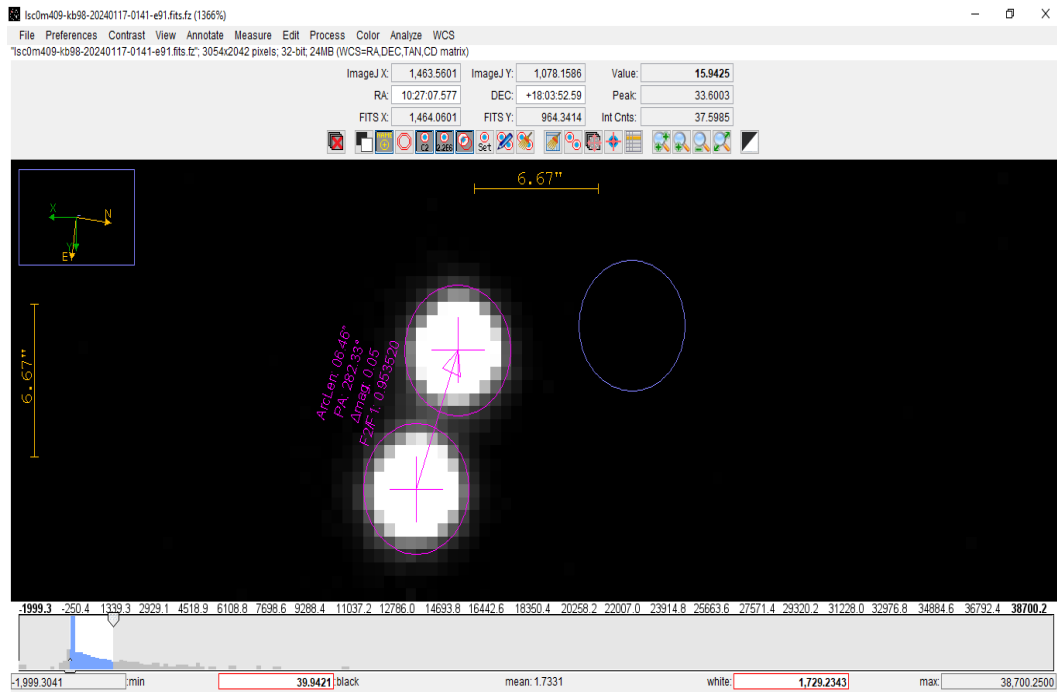


Figure 6: Data reduction process

5. Analysis Results

Newest measurement of the STF 1434 double stars was taken after the data reduction for the eight images by using AstroImageJ software. The results are shown in Table 1, while Table 2 showcases the statistical description of the measurements.

Table 1: Results of the measurements of 8 observations.

Label	Position_Angle(°)	Angular_Separation(")	Delta_Magnitude
lsc0m409-kb98-20240117-0141-e91.fits.fz	282.3270101	6.463592744	0.051675211
lsc0m409-kb98-20240117-0142-e91.fits.fz	282.1831061	6.474788499	0.039234419
lsc0m409-kb98-20240117-0143-e91.fits.fz	282.3279921	6.432073181	0.036788535
lsc0m409-kb98-20240117-0144-e91.fits.fz	282.154785	6.447276776	0.04259056
lsc0m409-kb98-20240117-0145-e91.fits.fz	282.0124771	6.538151427	0.051237206
lsc0m409-kb98-20240117-0146-e91.fits.fz	282.2513673	6.440618293	0.04042705
lsc0m409-kb98-20240117-0147-e91.fits.fz	282.3287948	6.432560316	0.038992201
lsc0m409-kb98-20240117-0148-e91.fits.fz	281.8042755	6.5002973	0.040847685

Table 2: Statistical Description of the measurements

	PA°	SEP/"	Delta_mag
Mean	282.173726	6.466169817	0.042724108
Standard Deviation	0.185283635	0.037280724	0.005641003
Standard Error of Mean	0.065507657	0.013180726	0.001994396

Figures 6, 7, and 8 provide evidence of the consistency of the newest measurements with the historical data obtained from USNO.

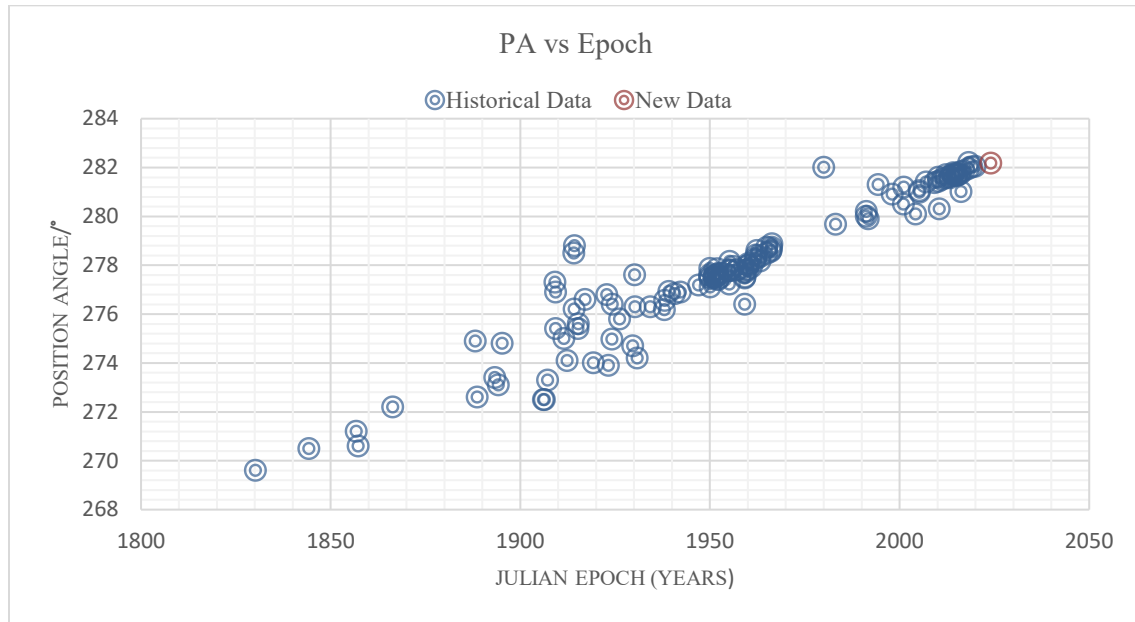


Figure 7: The graph of Position angle vs epoch

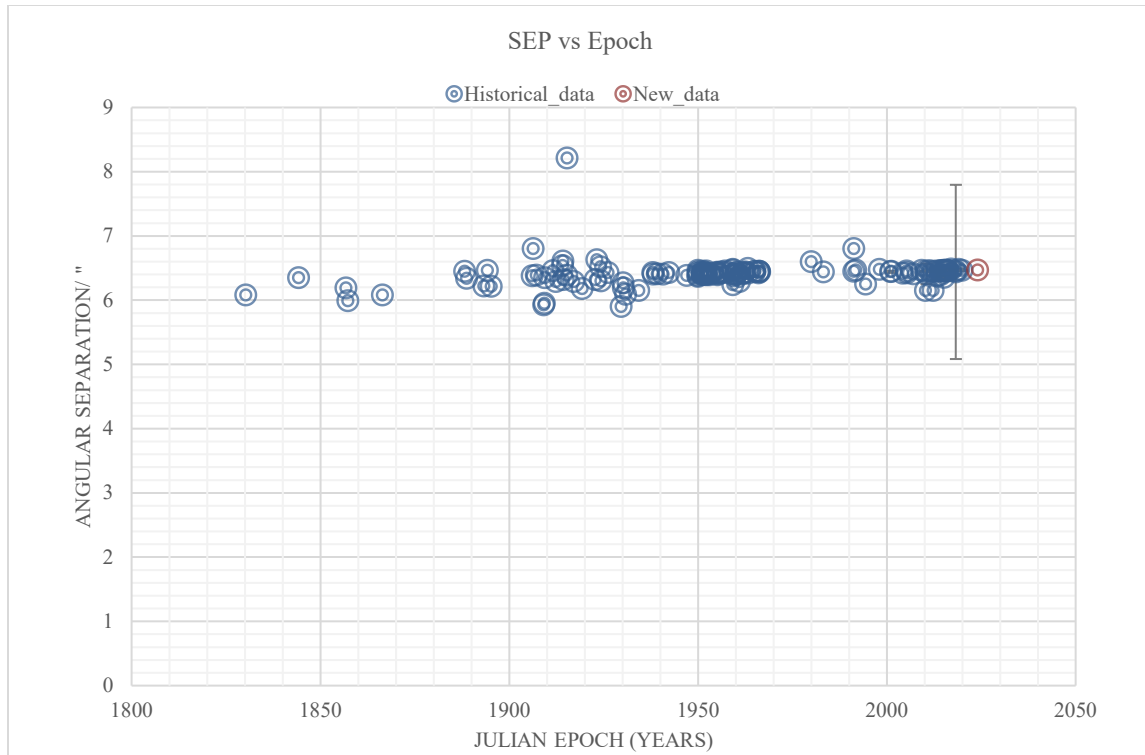


Figure 8: The graph of Angular Separation vs Epoch

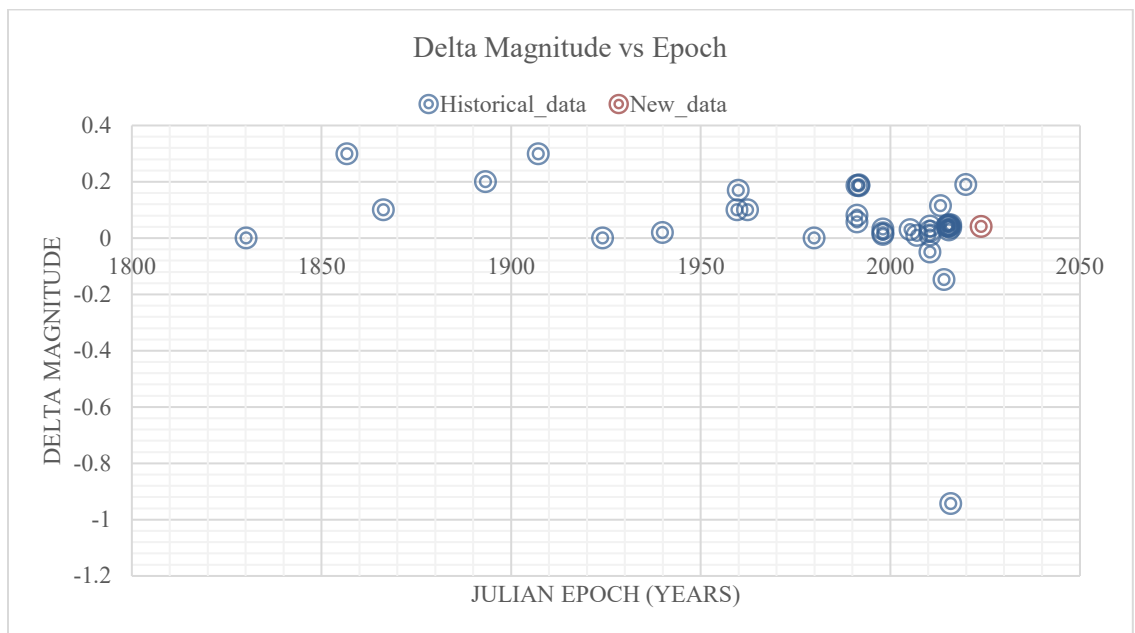


Figure 9: The graph of Delta Magnitude vs epoch

6. Conclusion

This study successfully took new measurements for the target double star STF 1434 (WDS 10274+1804). Measurements of Position Angle $282.1737^\circ \pm 0.0655^\circ$, Angular Separation $6.4661'' \pm 0.0131''$, and delta magnitude 0.0427 ± 0.0002 satisfied the aims of this study as it provides valuable insight into the future development of the field of double star astronomy. In this study, PA and SEP were consistent with the last observation, while the delta magnitude showed a significant change in the value compared to the last observation in 2019.

7. Acknowledgements

This research has made use of Las Cumbres Observatory 0.4 m telescope network as a part of 'Global Sky Partners' programme under the selected project 'LCOEPO2022B-007' Sri Lanka Astronomical Research Initiative (SLARI). The research team would like to thank Edward Gomez, Pat Boyce, Kalee Tok, Ryan Caputo and Rachel Freed for their guidance. Thanks are provided to the US Naval Observatory for their continued support of the double star programme, the Stelle Doppie catalogue maintained by Gianluca Sordi-glioni, and AstroImageJ software by Karen Collins and John Kielkopf at the University of Louisville. Finally, I would like to thank Prof. G.D.K. Mahanama and Dr. J.A.P. Bodhika at the Department of Physics, University Ruhuna, Sri Lanka.

References

- Badami, U. A., Tock, K., Carpenter, S., Kruger, K., Freed, R., Genet, R., Badami, U. A., Tock, K., Carpenter, S., Kruger, K., Freed, R., & Genet, R. (2018). Measurement of the Position Angle and Separation of HJ 1924. *JDSO*, 14(1), 181–189. <https://ui.adsabs.harvard.edu/abs/2018JDSO...14..181B/abstract>
- Cahill, J., Boysen, Z., Clark, M., Wagner, G., & Tock, K. (2018). *Comparison and CCD Measurements of Four Double Star Systems: WDS 03003+1432, 03009+5221, 03001+3911, 08165+7930* (Vol. 14, Issue 4).
- Dugan, O., Robinson, T., Carmeci, F., Tock, K., Dugan, O., Robinson, T., Carmeci, F., & Tock, K. (2019). CCD Measurements and Reclassification of WDS 07106 +1543 to an Optical Double. *JDSO*, 15(1), 119–129. <https://ui.adsabs.harvard.edu/abs/2019JDSO...15..119D/abstract>
- Dunlop, S. (2012). *Philip's Practical Astronomy* (2nd ed., Vol. 1). FIREFLY.
- Eggleton, P. (2006). *Evolutionary Processes in Binary and Multiple Stars*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511536205>
- Fraknoi, A., Morrison, D., & Wolff, S. C. (2016a). *Astronomy*. OpenStax. <https://openstax.org/details/books/astronomy>
- Fraknoi, A., Morrison, D., & Wolff, S. C. (2016b). *Astronomy*. <https://openstax.org/details/books/astronomy>